

Social Interactions and School Coexistence in Public Elementary Schools' Neurodiverse Classrooms

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- **Knowing how to live with others** who are different from us **is crucial** for satisfactory social relationships.
- Schools are a critical environment where all kinds of social interactions occur.
- Evidence has described **children with Autism Spectrum Disorder (ASD), and Attention Deficit and Hyperactivity Disorders (ADHD) displayed difficulties in peer functioning.**

1. **Children with autism and ADHD have lower centrality** in their peer social network compared to typically developing (TD) children and children with other special educational needs (SEN).
2. **Children with autism and ADHD engage in less reciprocal relationships** compared with TD children.

What is the role of autism and ADHD in social relationships in educational contexts?

What is a neurodiverse classrooms?

- In Chile students are assigned until graduation to a specific group of peers: the *classroom*.
- Neurodiverse classrooms are group of peers in which **children with special educational needs (SEN) engage and participate together** with other students in formative activities.



Experimental game theory



We designed a computational **non anonymous dictator game** that captures social interactions in an ecologically valid context.

1 Give informed consent and "earn" 15 stars

2 Select 10 classmates to play

3 Allocate their 15 stars to selected classmates

Javiera Juana Gonzalo Andrés Natalia

Pedro Roberto Mauricio Alonso Alfredo

Camila Olivia Fernando Juan Maureen

Daniela Natalia Roberto Mauricio Alonso

8/10

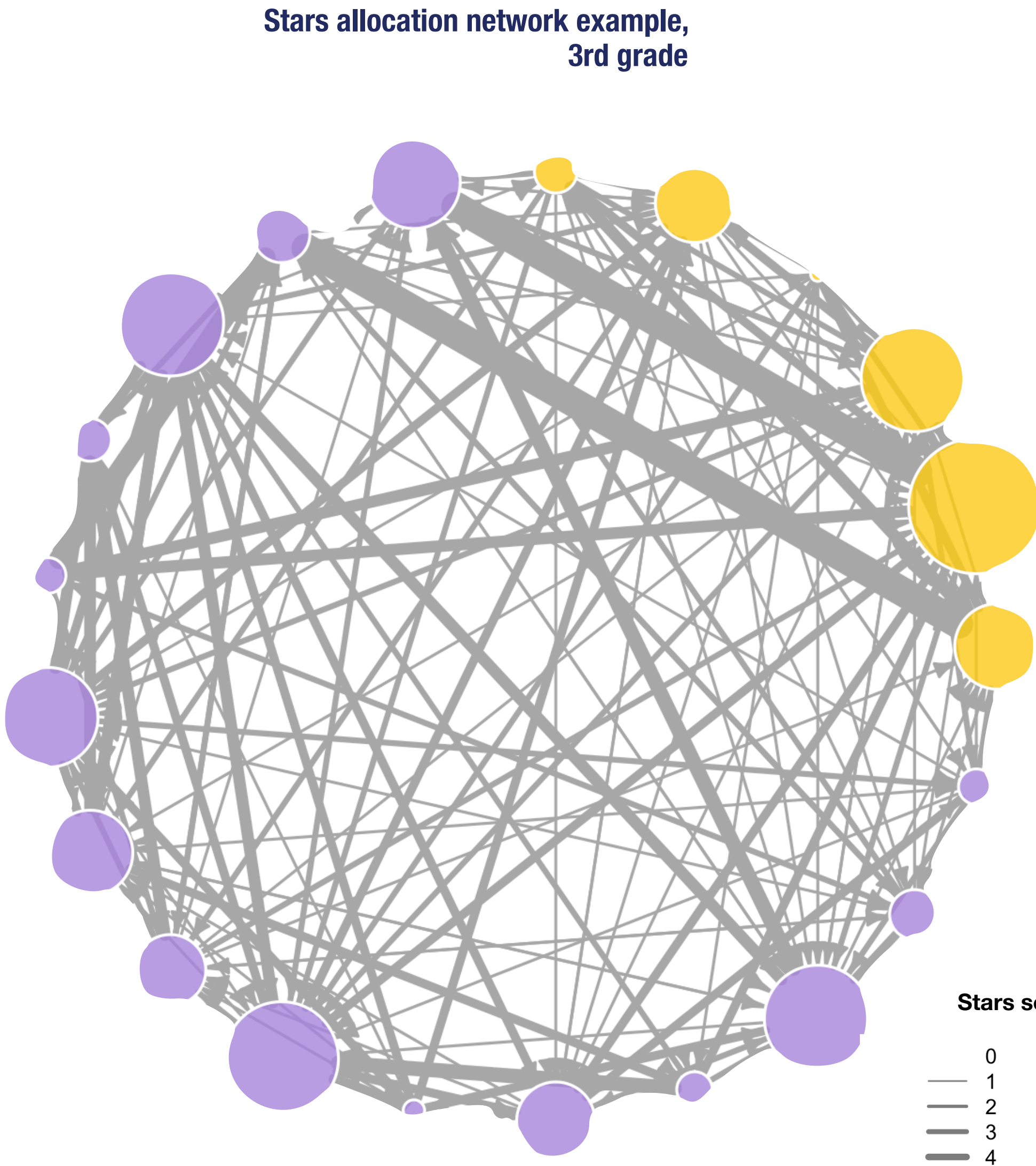
Watch a game demo

6 public schools in Chile 16 classrooms 705 enrolled students	625 Participants with legal guardians' informed consent	530 active players (sent and receive stars) 95 passive players (only receive stars)	28 Special educational needs
			488 typically developing
			35 Special educational needs
			95 typically developing



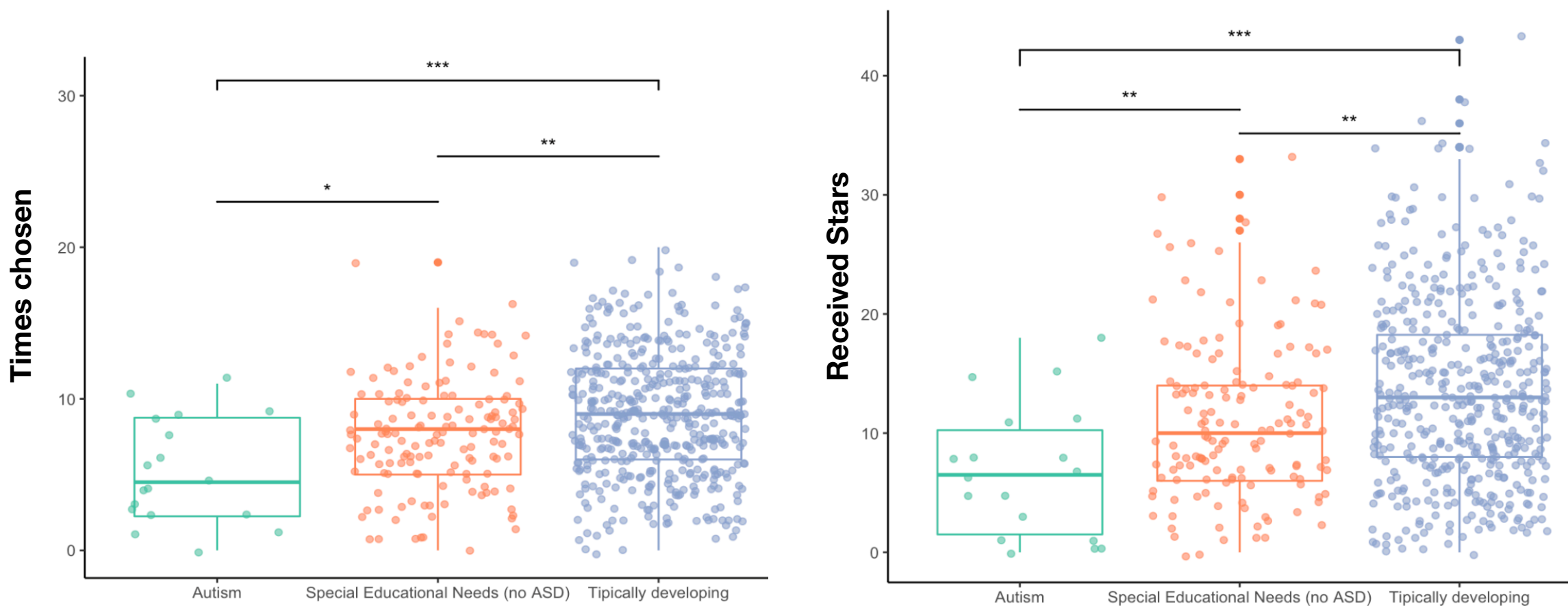
Social network analysis

We constructed **two interaction networks for each classroom** in which students are nodes, and links are defined by their behavior in the game.



- **Selection Network:**
 - Selection from participant A to B is a directed link: $A \rightarrow B$
 - We observe all choices for all players.
 - Centrality is the total of received selections.
- **Stars allocation Network:**
 - Is a subset conditional on selection.
 - Stars sent are weighted directed links.
 - Centrality is the sum of received stars.

1 Children with **special educational needs and autism have significantly lower centrality** compared to typically developing children.



2 Children with **autism significantly engage in less reciprocal relationships**, compared with typically developing children.

	Typically developing		Special educational needs (without considering Autism)		Autism	
	Chosen by A $A \rightarrow B$	Not chosen $A \rightarrow B$	Chosen by B $B \rightarrow A$	Not chosen $B \rightarrow A$	Chosen by B $B \rightarrow A$	Not chosen $B \rightarrow A$
Chosen by A $A \rightarrow B$	1685 (18.102%)	2088 (24.6%)	410 (18.102%)	535 (24.6%)	29 (13.3%)	28 (12.8%)
Not chosen $A \rightarrow B$	3276 (38.1%)	1558 (19.56%)	918 (38.1%)	503 (19.56%)	92 (38.3%)	69 (31.7%)

3 **Contrary to the literature**, children with ADHD did not show any significant differences to both TD children and children with special educational needs, including autism.

Explore more of our results



Discussion

Certain neurodevelopment conditions are related to the formation of social interaction networks in elementary classrooms and its impact on school coexistence.

Understanding social interactions in neurodiverse groups can contribute to plan policies that improve school integration and coexistence.

References

